Run It, Then Cite It: Execution- and Mutation-Verified Answers to Developer Questions

ABSTRACT

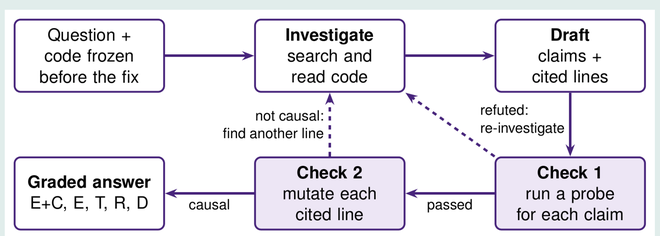
Developers often ask how a library really behaves when its documentation may be wrong. LLM assistants tend to repeat the docs and cite code that looks related but does not decide the behaviour. We built an agent that checks its answer twice: it runs a small test (a probe) for every claim, and it keeps a cited line only if changing that line changes the test's result. On 145 real GitHub issues from click and marshmallow, with pre-registered hypotheses, it did not beat a ReAct agent with the same tools. The line-changing check turned out to be the weak part.

METHODOLOGY

Data. 170 questions built like SWE-bench: each is a GitHub issue, and its merged fix and test are the answer key. Code is frozen before the fix; no LLM writes the gold. A human spot check left 145 test questions.

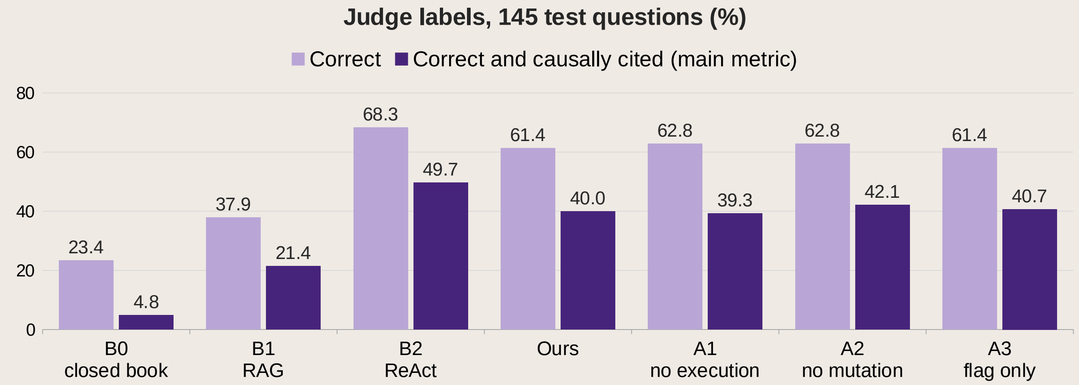
Agent. Drafts claims with cited lines. Check 1 runs a probe per claim in an offline Docker sandbox. Check 2 mutates each cited line (e.g. 100→101, < → <=) and accepts it only if the claim flips while control probes stay the same.

Baselines. B0 closed book, B1 one-call RAG, B2 ReAct with all the same tools and budget.



RESULTS

On 145 locked test questions, our agent did not beat ReAct (B2) with the same tools: 40.0% vs 49.7% correct and causally cited answers by the LLM judge, and 25.0% vs 41.7% on 12 human-labelled questions. Both tool-using agents beat closed-book and RAG by a wide margin.



HYPOTHESES

Written down before any test run. The LLM judge agreed with humans only weakly (kappa 0.36 and 0.32, below our 0.6 bar), so human labels decide. None of the three held.

	Prediction → result
H1	Ours beats B2 by ≥ 8 points → No: 25.0% vs 41.7% (humans), 40.0% vs 49.7% (judge)
H2	40% fewer confident errors → No: 29.7% vs 31.7%
H3	Mutation raises citation precision by ≥ 15 points → Opposite: 30.6% with it, 44.1% without

WHY IT FAILED

Check 2 throws away good lines. The rule we used rejected 76% of the lines developers actually fixed, and its verdicts agree with humans no better than chance. In click-2099 it dropped the developer's own fix line.

Mutation rule	Fixed lines accepted	Kappa vs humans
any change	66.8%	0.20
not just breakage	56.1%	0.17
+ controls unchanged (used)	24.0%	−0.10

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CONCLUSION AND FUTURE WORK

- In a fair, pre-registered test, checking by execution and mutation did not beat a simpler ReAct agent.
- Flagging failed checks, instead of re-investigating them, cut confident errors by a third.
- No unsafe action ran, but planted text could still change answers.
- Next: a softer causal test that ranks lines instead of a yes/no filter; run Chain-of-Verification and a larger-budget ReAct.

ACKNOWLEDGMENT

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REFERENCES

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Code, data and full report

github.com/ameya-purohit/LMA-Final-Project

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